

Ali Hatem Tahssin

Computer and AI Engineering Student

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Education

Ain Shams University (iCHEP)

Computer and Artificial Intelligence Engineering | Sep 2023 – Present | Graduation: 2027

Cairo, Egypt

University of East London (Dual Degree)

Computer and Artificial Intelligence Engineering | Sep 2023 – Present | Graduation: 2027

Cairo, Egypt

Training & Certification

Digital Twin Technologies by Siemens

April 2026

- Covered the 5D digital twin lifecycle, virtual product design and validation, and smart system architectures using the Innexis Virtual System Interconnect framework

Data Science Training by CLS Learning

Nov 2025

- Covered Python, SQL, statistics and probability, and linear algebra; trained supervised and unsupervised models
- Dived into deep learning, NLP, Gen AI, and MLOps
- Built data pipelines with NumPy, Pandas, and scikit-learn including preprocessing, feature engineering, SMOTE for class imbalance, and Stratified K-Fold cross-validation

Linux Administration by Edges for Training

Aug 2025

- Mastered Linux CLI, advanced Bash scripting, user/group permission management, process and package management, networking, services and systemd, disks and filesystems, and Makefile/CMake build systems

Competitive Programming Summer Training by Coach Academy

Aug 2024

- Intensive training in algorithm design, complexity analysis, STL containers, dynamic programming, and recursion; competed in an on-site programming contest

Projects

Async Health Monitor System | Python

- Built an async health-monitoring backend in FastAPI with per-service check loops and PostgreSQL storage via SQLAlchemy
- Containerized the app with Docker and Compose for local dev and production, using an override file to avoid touching the prod config
- Provisioned AWS EC2 infrastructure with Terraform, including networking and automated Docker install on boot.
- Wrote a 4-layer test suite (unit, integration, e2e, Postman) with a 4-stage GitHub Actions CI/CD pipeline: test, build, push, deploy.

Distributed LLM Inference System | Python

- Architected a fault-tolerant master-worker system on AWS EC2, using NGINX to load-balance requests into a FastAPI scheduler dispatching inference across 3 worker nodes
- Built 3 load-balancing strategies, zero-downtime hot-swapping, heartbeat failover, and per-worker semaphores to prevent OOM under load
- Engineered a hybrid RAG pipeline (ChromaDB and keyword fallback) feeding a built in Ollama LLM
- Load-tested across 8 waves scaling 10 to 100 concurrent users, hitting 100% success rate at sub-8.2s P95 latency; built a live dashboard for per-worker CPU/memory/connection metrics with 2s refresh

Heart Disease Predictor | Python

- Built an end-to-end data mining pipeline for heart disease prediction, covering data cleaning, missing-value handling, outlier detection, categorical encoding, exploratory analysis, and 80/20 train-test splitting
- Implemented and evaluated 5 classification models: Logistic Regression, Decision Tree, Random Forest, KNN, and Naive Bayes, using accuracy, precision, recall, F1-score, error rate, and confusion matrices
- Compared model performance and feature importance, achieving 91.8% test accuracy with Logistic Regression

Road Accident Severity Predictor | Python

- Built a multi-class ML pipeline to predict Slight, Serious, and Fatal accident severity using temporal, road, weather, and lighting features
- Trained and compared 5 classification models with SMOTE and 5-fold stratified cross-validation, selecting the best model by macro-F1
- Generated confusion matrices, feature importance, and error analysis, and serialized the final prediction pipeline

Technical Skills

- AI & Machine Learning:** scikit-learn, PyTorch, pandas, NumPy, EDA, Feature Engineering, Model Evaluation
- Gen AI:** LLMs, RAG, Prompt Engineering, Embeddings, Vector Databases, LLM APIs
- Backend & APIs:** FastAPI, Node.js, Express, REST API, JWT, async/await, SQLAlchemy, Mongoose, Pydantic
- Databases:** PostgreSQL, MongoDB, SQL, Redis
- Programming Languages:** Python, JavaScript, TypeScript, Java, C++, C#
- Cloud & DevOps:** Git, GitHub, Docker, CI/CD, GitHub Actions, Linux, Terraform, AWS, EC2, S3, AMI
- Testing:** Pytest, Postman
- Frontend:** React, HTML, CSS
- Languages:** Arabic (native), English (fluent)